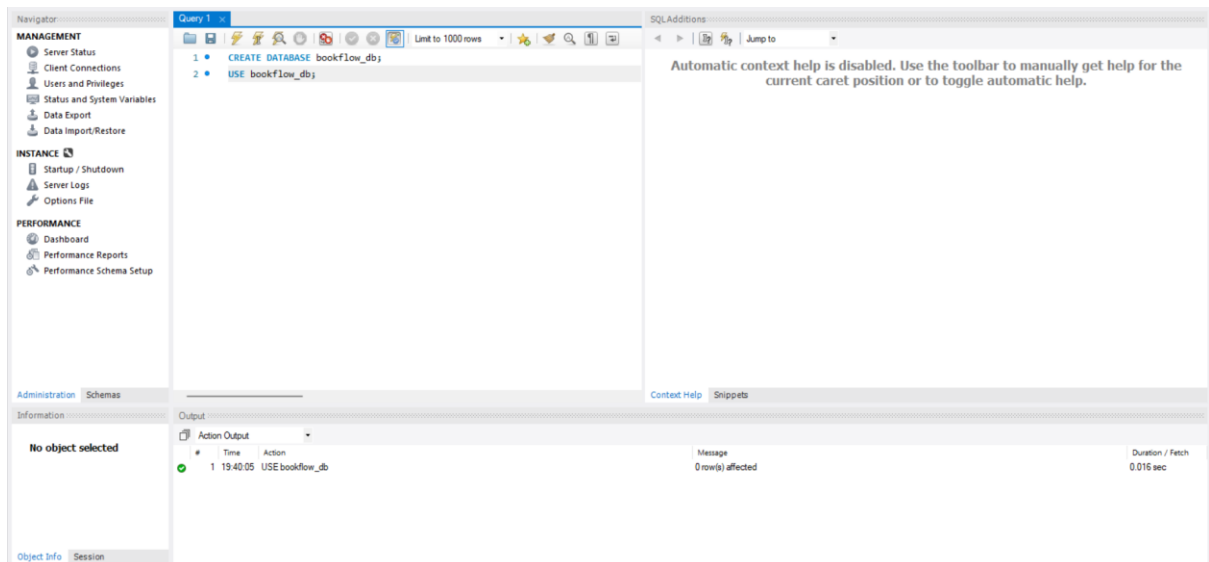
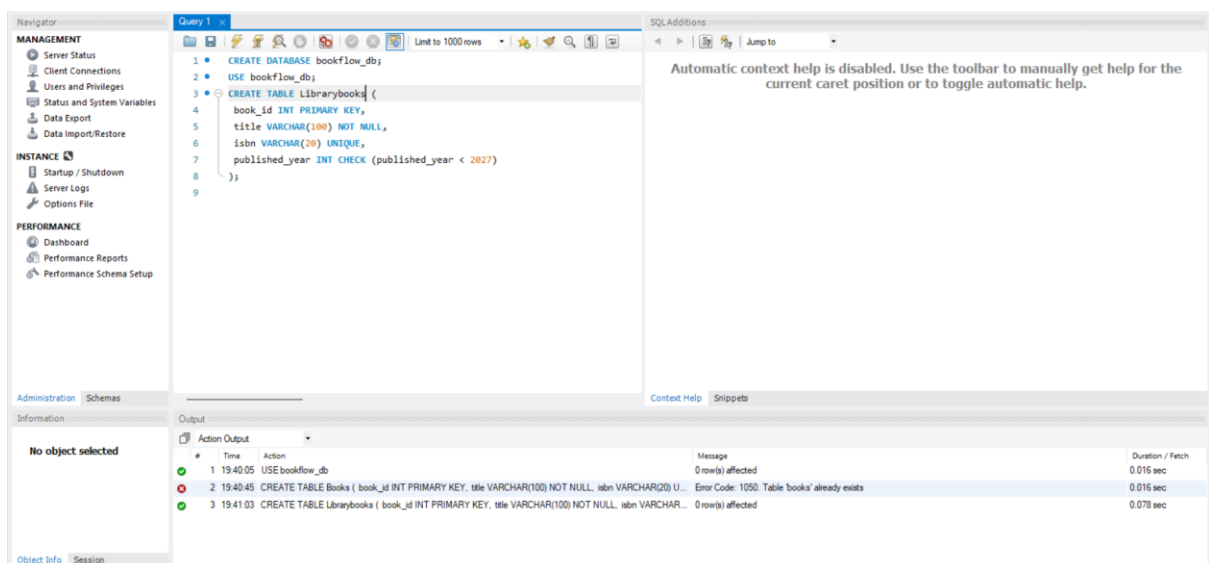


WEEK-2

Step 1 — Create Database



Step 2 — Create Books Table (DDL + Constraints)



Step 3 & 4 — Insert Books and Display

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

```

7  published_year INT CHECK (published_year < 2027)
8  );
9  • INSERT INTO Books (book_id, title, isbn, published_year) VALUES
10 (1, 'The Great Gatsby', '9780743273565', 1925),
11 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
12 (3, '1984', '9780451524935', 1949);
13 • SELECT * FROM Books;

```

Result Grid

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

Books 3 x

Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:40:05	USE bookflow_db	0 row(s) affected	0.016 sec
2	19:40:45	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) ...	Error Code: 1050. Table 'books' already exists	0.016 sec
3	19:41:03	CREATE TABLE Librarybooks (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	0 row(s) affected	0.078 sec
4	19:41:47	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.015 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Step 5, 6 & 7 — Members Table: Create, Insert, Display

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

```

17 email VARCHAR(100) UNIQUE
18 );
19 • INSERT INTO Members (member_id, full_name, email) VALUES
20 (101, 'John Smith', 'john.smith@email.com'),
21 (102, 'Emma Wilson', 'emma.wilson@email.com'),
22 (103, 'Michael Brown', 'michael.brown@email.com');
23 • SELECT * FROM Members;

```

Result Grid

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com

Members 4 x

Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:40:05	USE bookflow_db	0 row(s) affected	0.016 sec
2	19:40:45	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) ...	Error Code: 1050. Table 'books' already exists	0.016 sec
3	19:41:03	CREATE TABLE Librarybooks (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	0 row(s) affected	0.078 sec
4	19:41:47	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.015 sec / 0.000 sec
5	19:43:05	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Step 8 — Create Loans Table (FOREIGN KEYS)

Query 1

```

13 SELECT * FROM Books;
14 CREATE TABLE Members (
15     member_id INT PRIMARY KEY,
16     full_name VARCHAR(100),
17     email VARCHAR(100) UNIQUE
18 );
19 INSERT INTO Members (member_id, full_name, email) VALUES
20     (101, 'John Smith', 'john.smith@email.com'),
21     (102, 'Emma Wilson', 'emma.wilson@email.com'),
22     (103, 'Michael Brown', 'michael.brown@email.com');
23 SELECT * FROM Members;
24 CREATE TABLE Loans (
25     loan_id INT PRIMARY KEY,
26     member_id INT,
27     book_id INT,
28     loan_date DATE,
29     FOREIGN KEY (member_id) REFERENCES Members(member_id),
30     FOREIGN KEY (book_id) REFERENCES Books(book_id)
31 );
32

```

Output

#	Time	Action	Message	Duration / Fetch
1	19:40:05	USE bookflow_db	0 row(s) affected	0.016 sec
2	19:40:45	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) ...	Error Code: 1050. Table 'books' already exists	0.016 sec
3	19:41:03	CREATE TABLE Librarybooks (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHA...	0 row(s) affected	0.078 sec
4	19:41:47	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.015 sec / 0.000 sec
5	19:43:05	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
6	19:43:55	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FORE...	0 row(s) affected	0.062 sec

Step 9 & 10 — Insert 10 Loans and Display

Query 1

```

32 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
33     (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
34     (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
35     (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
36     (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
37     (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
38 SELECT * FROM Loans;

```

Output

#	Time	Action	Message	Duration / Fetch
2	19:40:45	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20)...	Error Code: 1050. Table 'books' already exists	0.016 sec
3	19:41:03	CREATE TABLE Librarybooks (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHA...	0 row(s) affected	0.078 sec
4	19:41:47	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.015 sec / 0.000 sec
5	19:43:05	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
6	19:43:55	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, PO...	0 row(s) affected	0.062 sec
7	19:44:30	SELECT * FROM Loans LIMIT 0.1000	0 row(s) returned	0.032 sec / 0.000 sec

Step 11 — JOIN Query (Who Borrowed What)

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

```

36 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
37 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
38 SELECT * FROM Loans;
39 SELECT m.full_name AS Member_Name, b.title AS Book_Title
40 FROM Loans l
41 INNER JOIN Members m ON l.member_id = m.member_id
42 INNER JOIN Books b ON l.book_id = b.book_id;

```

Result Grid

Member_Name	Book_Title

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	19:41:03	CREATE TABLE Librarybooks (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCH...	0 row(s) affected	0.078 sec
4	19:41:47	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.015 sec / 0.000 sec
5	19:43:05	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
6	19:43:55	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FO...	0 row(s) affected	0.062 sec
7	19:44:30	SELECT * FROM Loans LIMIT 0.1000	0 row(s) returned	0.032 sec / 0.000 sec
8	19:45:06	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.m...	0 row(s) returned	0.000 sec / 0.000 sec

Step 12 — GROUP BY Query (Books per Year)

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

```

40 FROM Loans l
41 INNER JOIN Members m ON l.member_id = m.member_id
42 INNER JOIN Books b ON l.book_id = b.book_id;
43 SELECT published_year, COUNT(book_id) AS Total_Books
44 FROM Books
45 GROUP BY published_year;
46 ORDER BY published_year;

```

Result Grid

published_year	Total_Books
1988	1
2008	1
2018	1

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	19:41:47	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.015 sec / 0.000 sec
5	19:43:05	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
6	19:43:55	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FO...	0 row(s) affected	0.062 sec
7	19:44:30	SELECT * FROM Loans LIMIT 0.1000	0 row(s) returned	0.032 sec / 0.000 sec
8	19:45:06	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.m...	0 row(s) returned	0.000 sec / 0.000 sec
9	19:45:35	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDE...	3 row(s) returned	0.016 sec / 0.000 sec

Step 13-16 — Donation via TRANSACTION (Atomicity)

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

```

41 INNER JOIN Members m ON l.member_id = m.member_id
42 INNER JOIN Books b ON l.book_id = b.book_id;
43 SELECT published_year, COUNT(book_id) AS Total_Books
44 FROM Books
45 GROUP BY published_year;
46 ORDER BY published_year;
47 CREATE TABLE Donation_History (
48     donation_id INT PRIMARY KEY,
49     book_id INT,
50     donor_name VARCHAR(100),
51     donation_date DATE,
52     FOREIGN KEY (book_id) REFERENCES Books(book_id)
53 );
54 START TRANSACTION;
55 INSERT INTO Books (book_id, title, isbn, published_year)
56 VALUES (4, 'Animal Farm', '9780451526342', 1945);
57 INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
58 VALUES (1, 4, 'Raj Kumar', CURDATE());
59 COMMIT;
60

```

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	19:43:05	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
6	19:43:55	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FO...	0 row(s) affected	0.062 sec
7	19:44:30	SELECT * FROM Loans LIMIT 0.1000	0 row(s) returned	0.032 sec / 0.000 sec
8	19:45:06	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.m...	0 row(s) returned	0.000 sec / 0.000 sec
9	19:45:35	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDE...	3 row(s) returned	0.016 sec / 0.000 sec
10	19:46:12	COMMIT	0 row(s) affected	0.000 sec

Step 17 & 18 — Index on ISBN + Fast Search

The screenshot displays the SQL Developer interface. The main window shows a query titled "Query 1" with the following SQL code:

```
55 • INSERT INTO Books (book_id, title, isbn, published_year)
56 VALUES (4, 'Animal Farm', '9780451526342', 1945);
57 • INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_
58 VALUES (1, 4, 'Raj Kumar', CURDATE());
59 • COMMIT;
60 • CREATE INDEX idx_books_isbn ON Books(isbn);
61 • SELECT * FROM Books WHERE isbn = '9780451524935';
```

The output window shows the execution results for the query. The output is as follows:

#	Time	Action	Message	Duration / Fetch
6	19:43:55	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, PO...	0 row(s) affected	0.062 sec
7	19:44:30	SELECT * FROM Loans LIMIT 0, 1000	0 row(s) returned	0.032 sec / 0.000 sec
8	19:45:06	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.m...	0 row(s) returned	0.000 sec / 0.000 sec
9	19:45:35	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDE...	3 row(s) returned	0.016 sec / 0.000 sec
10	19:46:12	COMMIT	0 row(s) affected	0.000 sec
11	19:47:45	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec